Week 7: “Behavioral” Labor Supply

ECON 6290
Lisa Ho
Wednesday, October 21, 2026

Announcements

1. Mini-proposal #1: feedback back to you by the end of this week (Friday, October 23rd)

2. Mini-proposal #2 is due Friday, November 6th

3. Shape of the day
   - Identity (Oh 2023)
   - Financial strain and productivity (Kaur, Mullainathan, Oh, and Schilbach 2025)
   - Break (5 minutes)
   - Discussion: Hussam, Kelley, Lane, and Zahra, “The psychosocial value of employment” (Emilio leading)
   - Co-workers’ pay and morale (Breza, Kaur, and Shamdasani 2018)

4. Any questions?

Four components of labor supply that standard models leave out

- Standard labor supply: a worker chooses hours and effort to trade off consumption against leisure and the disutility of effort, given their own wage and non-labor income

- Today: four components of labor supply that this model does not traditionally include
  1. The worker's own identity and (in)congruence with the job tasks
  2. Psychological stress: how financial strain can itself lower job performance
  3. The psychosocial value of being employed
  4. Relative pay: morale depends on how pay compares to those of co-workers

“Behavioral” labor supply

1. **“Does Identity Affect Labor Supply?”**
   Oh (2023), *American Economic Review*
2. **“Do Financial Concerns Make Workers Less Productive?”**
   Kaur, Mullainathan, Oh, and Schilbach (2025), *Quarterly Journal of Economics*
3. **“The Psychosocial Value of Employment: Evidence from a Refugee Camp”**
   Hussam, Kelley, Lane, and Zahra (2022), *American Economic Review* (discussion lead: Emilio)
4. **“The Morale Effects of Pay Inequality”**
   Breza, Kaur, and Shamdasani (2018), *Quarterly Journal of Economics*

Does Identity Affect Labor Supply? (Oh, 2023)

- Identity refers to individuals' concepts of "who they are"
  - Often oriented around social categories (e.g., women, parents)
  - These social categories may be associated with behavioral prescriptions, dictating how the members of the category should act

- Violating such prescriptions could lead to emotional or psychological costs (Akerlof & Kranton 2000)
  - Protecting identity may compel someone to follow internal rules of behavior
  - (Even when no one can observe her or when it is economically costly to do so)
  - Overviews of identity theories from social psychology: Stryker (1982), Burke & Stets (2009)
    From economics: Hoff & Stiglitz (2016), Shayo (2020)

In this study, focus on **caste** as relevant aspect of identity

- Caste is inherited at birth: about 4,000 endogamous communities (*jatis*) (Munshi 2019)
  - Ranked by ritual purity: four *varnas* (Brahmins, Kshatriyas, Vaishyas, Shudras), with the former "untouchables" outside and below them (Deshpande 2011)
  - Official categories today: forward castes, Other Backward Classes (OBC), Scheduled Castes (SC, 16.6% of the population), and Scheduled Tribes (ST, 8.6%) (Census of India 2011)

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- Caste traditionally fixed occupation as well as status
  - Being born into a caste meant being born into its occupation, e.g., the Dhoba caste in Odisha and washing clothes (Shah et al. 2006)
  - Today, workers are 3$\times$ as likely to work in their caste's traditional occupation as random assignment implies (Cassan, Keniston, and Kleineberg 2022)

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  - Today, workers are 3× as likely to work in their caste's traditional occupation as random assignment implies (Cassan, Keniston, and Kleineberg 2022)
- Caste norms still govern daily life
  - Only 5% of married women report an inter-caste marriage (India Human Development Survey, IHDS 2011–12)
  - 27% of households report practicing untouchability, e.g., not letting an SC person enter the kitchen (IHDS 2011–12)

Setting: casual laborers in rural Odisha

- Lean seasons of October–November 2018 and June–August 2019
- Main experiment: 630 men from 141 villages, 7 castes (6 Scheduled Castes, 1 OBC)
  - Male household heads aged 18–55
  - Work regularly as casual wage laborers, i.e., not mainly in a caste occupation
  - About 2.5 days of paid work in the previous week

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  - Work regularly as casual wage laborers, i.e., not mainly in a caste occupation
  - About 2.5 days of paid work in the previous week
- Two background surveys, in separate villages
  - Task Survey ($N = 151$): which castes perform which manual tasks
  - Rank Survey ($N = 209$): card-sort the 7 castes into a hierarchy
- Supplementary bonus experiment: 106 men from 2 castes, 12 villages

Castes and caste tasks in the study; paired tasks developed as “twins”

Caste	Rank score (1 = top)
Kaibarta	1.48
Sundhi	2.07
Dhoba	3.71
Kela	4.14
Mochi	4.59
Pana	5.19
Hadi	6.60

Castes and caste tasks in the study; paired tasks developed as “twins”

Caste	Rank score (1 = top)	Identity task	% linking to caste
Kaibarta	1.48	–	
Sundhi	2.07	–	
Dhoba	3.71	Washing clothes	72
Kela	4.14	–	
Mochi	4.59	Mending leather shoes	97
Pana	5.19	–	
Hadi	6.60	Sweeping latrines	84

Castes and caste tasks in the study; paired tasks developed as “twins”

Caste	Rank score (1 = top)	Identity task	% linking to caste	Paired control task	% linking to any SC
Kaibarta	1.48	–		–	
Sundhi	2.07	–		–	
Dhoba	3.71	Washing clothes	72	Washing farming tools	0
Kela	4.14	–		–	
Mochi	4.59	Mending leather shoes	97	Mending grass mats	15
Pana	5.19	–		–	
Hadi	6.60	Sweeping latrines	84	Sweeping animal sheds	0

- What makes each one a control for its identity task:
  1. Same physical action and skill: washing, mending, or sweeping
  2. No caste association: no one links it to the paired caste (at most 15% to any SC)
  3. No strong association with women’s work

Elicit willingness to do caste vs not-caste-associated task

- Offer 1-day jobs that differ only in an *extra* task which is caste or not caste associated
  - Core job pays Rs 300 for 5 hours of work in making paper bags
  - 8 extra tasks (3 caste-associated, 3 paired control, 2 pure control)
  - Also vary duration of extra task - 4 durations: 10, 30, 60, or 90 minutes
  - Extra task done in a private space far from home; tasks don't require experience to do

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- Elicitation: incentive-compatible "take-it-or-leave-it" set of decisions over job offers
  1. For a list of job offers, consider each offer and decide whether to accept or decline
  2. Then one job offer is randomly drawn & choice is implemented
     - If said "yes" to job offer, then job is offered and participant is supposed to do the job
     - If said "no" to job offer, then nothing happens (no job from the employer)

Aside: techniques for WTP elicitations

1. Strategy method (like in this paper)
   - Any elicitation where you make a choice over a series of decisions, then one decision is randomly drawn and implemented

2. Multiple price list (MPL)
   - Series of decisions that vary in price, you say yes/no to each of them, then one row is randomly drawn and implemented

3. Becker-DeGroot-Marschak (BDM)
   - State a reservation price, then a random price is drawn $\rightarrow$ you buy if your reservation price is above drawn price
   - Pro is that you get more detail on exactly someone's willingness to pay
   - Con is that it's harder for participants to understand

Example of a multiple price list

Would you buy this solar lamp at this price?

Price	Buy?
$1	**Yes**
$2	**Yes**
$3	**Yes**
$4	**Yes**
$5	No
$6	No
$8	No
$10	No

- The respondent answers every row before anything happens

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- The respondent answers every row before anything happens
- Then one row is drawn at random and that choice is carried out
  - $3 drawn: pays $3 and takes the lamp home
  - $6 drawn: no sale
- What can you infer about WTP from these responses?

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  - $\text{WTP} \geq \$10$

Example of the Becker-DeGroot-Marschak (BDM) mechanism

1. First, ask the respondent what the most is that they are willing to pay for the solar lamp.
    - Suppose they say $4
2. A price *p* is drawn at random, e.g., using a large blue 10-sided dice ($0 to $10)
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- Suppose Lisa says that she has figured out how to game the system. She values the solar lamp at $4 but will say $3 “to get a better deal.” Why is this a bad idea?

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Stating her true WTP is optimal: the bid decides *whether* she buys, never *what* she pays
In practice though BDM can be a pain to implement.

Job offer response sheet in Oh (2023)

Response Entry – Job Offers

Time		10 minutes	30 minutes	1 hour	1 hour and 30 minutes
					
Task					
1					
2					
3					
4					
5					
6					
7					
8					

- One cell per job offer: 8 extra tasks $\times$ 4 durations = 32 offers
- Tasks and durations are shown as pictures
- The surveyor describes each task with photos, then marks ✓ (accept) or ✗ (decline) in every cell

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- Tasks and durations are shown as pictures
- The surveyor describes each task with photos, then marks ✓ (accept) or ✗ (decline) in every cell
- Then the worker rolls an 8-sided die (task) and picks a scratch card (duration)
- Only that one offer is implemented

How the script explains the choices (can find these things in the appendix)

We will go through this list later, and you will choose whether you prefer to accept or decline each job offer. For every offer, we will mark a (✓) if you would like to accept the job offer, and mark a (X) if you would like to decline the job offer. Accepting any offer means that if this particular offer is randomly chosen, you agree to working on the specified extra task first to fulfil the requirement, then working on paper bags for the remaining time, and receiving [insert wage] at the end of the day. Declining any offer means if this particular offer is chosen, you prefer to not work at all with us and will not earn any wages from us. Remember, you will only get one offer, so it is important you think about each offer carefully.

Now let me tell you how the dice roll process determines your job offer. You die roll determines which extra task you have to work on, as part of your job. A scratch card will determine how much you are required to work on the extra task. These two together will determine the exact type of your job offer. As in the last exercise, because your job offer is determined by chance, just picking your honest preferred choice for every offer is the best thing you can do for yourself.

Remember, if you accepted any offer and got it, you cannot later refuse to work on the extra task. If you do not like the idea of working on the extra task for the requirement amount, you should just decline the offer. That is, say you agree to work on a certain task to complete the requirement. You roll the dice and that is the option you get. You cannot change your answer at this point. We cannot offer you any other job or allow you to re-roll the dice. If you do not complete the job, we will come back to your village and ask you to complete the job. So please only agree to a job if you are comfortable performing it [today or tomorrow].

If you are okay with working on the extra task for the required amount, then you should always accept the offer. That is, say yes to a job offer if you are comfortable performing it. If this offer gets randomly chosen, you will get work [today or tomorrow] and get [insert wage] at the end of the day.

Oh's survey instrument: practice round and comprehension checks

First a practice round with real stakes: Rs 10 food packets (tea, sugar, lentils, spices)

Since your sale offer is determined by chance, when you are deciding about each offer, all you need to think about is whether you want to take this offer or not. I am saying this, because sometimes people mistake this exercise as a bargaining situation. If we were bargaining, even if you like some offer, you may say you do not like this offer so that I give you a lower price. However, in this exercise, there is no bargaining. The dice roll and scratch cards randomly give you only one offer, and this offer does not change. Therefore, you can just give a simple honest answer, since all you need to consider is whether you want to buy offered products for Rs 10, or not buy them. Do you understand?

Then comprehension questions (up to 3 tries) before the real job offers

Now let's just make sure you completely understand the process.

C1. What does the die roll determine?

Try #1: ☐ Yes ☐ No Try #2: ☐ Yes ☐ No Try #3: ☐ Yes ☐ No

C2. What does the scratch card determine?

Try #1: ☐ Yes ☐ No Try #2: ☐ Yes ☐ No Try #3: ☐ Yes ☐ No

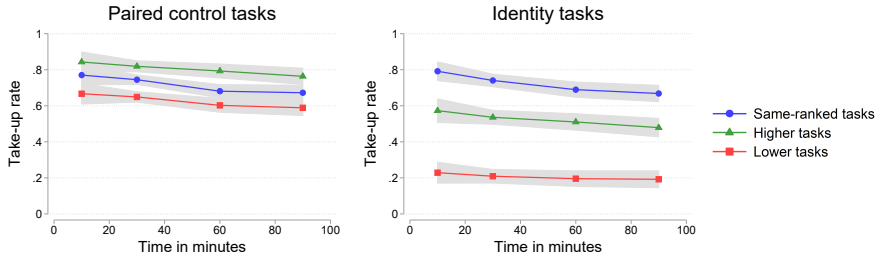
C3. What happens if you get an offer, and if you marked (✓) for this option?

Try #1: ☐ Yes ☐ No Try #2: ☐ Yes ☐ No Try #3: ☐ Yes ☐ No

C4. What happens if you get an offer, and if you marked (X) for this option?

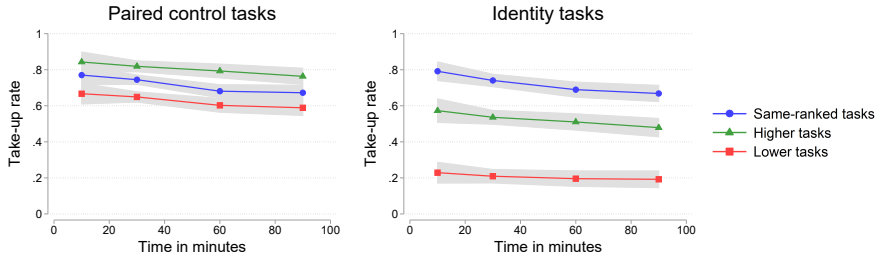
Try #1: ☐ Yes ☐ No Try #2: ☐ Yes ☐ No Try #3: ☐ Yes ☐ No

Raw take-up



- Same-ranked (blue) identity tasks match their pairs. Higher (green) and lower (red) ones are shifted down at every duration

Raw take-up



- Same-ranked (blue) identity tasks match their pairs. Higher (green) and lower (red) ones are shifted down at every duration
- Why are paired control tasks different in levels? My guess is b/c each color is different workers
  - Green (higher tasks): mostly low-ranked, poorer castes, who take more jobs of any kind
  - Red (lower tasks): mostly the richer Kaibarta and Sundhi (34 pp more likely to own land)

Estimate the "identity effect" as a difference in differences

$$Y_{ikt} = \sigma^d D_{ik} + \lambda^d \, \text{Id}_k D_{ik} + \sigma^l L_{ik} + \lambda^l \, \text{Id}_k L_{ik} + \tau \cdot \text{hours}_t + \text{task FE} + \text{worker FE} + \varepsilon_{ikt}$$

- $Y_{ikt} = 1$ if worker i accepts the offer with task k for time t.
- $\text{Id}_k = 1$ for an identity task, 0 for its paired control
- $D_{ik} = 1$ if task k's caste is ranked differently from worker i's caste. $L_{ik} = 1$ if it is ranked lower

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- What each control is doing:
  - Task FE: how each task differs from its pair for everyone (e.g. dirtier)
  - Worker FE: who the worker is (wealth, outside options)
  - σ^d, σ^l: how take-up of the paired controls varies with rank, where no identity is at stake (picks up anything about rank that affects the identity task and its pair equally, e.g. caste A likes or dislikes a particular kind of work)
- The λ's are the parameters of interest.
  - What is λ^d?

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  - What is λ^l?

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- λ^d: how much lower take-up of an identity task is, relative to its pair, when the task belongs to a higher-ranked caste instead of the worker's own caste
- λ^l: how much further that gap falls when the task belongs to a lower-ranked caste instead of a higher-ranked one

$$\lambda^d = \left(\bar{Y}^{\mathrm{Id}} - \bar{Y}^{\mathrm{pair}}\right)_{\text{higher}} - \left(\bar{Y}^{\mathrm{Id}} - \bar{Y}^{\mathrm{pair}}\right)_{\text{same-ranked}}$$

$$\lambda^d + \lambda^l = \left(\bar{Y}^{\mathrm{Id}} - \bar{Y}^{\mathrm{pair}}\right)_{\text{lower}} - \left(\bar{Y}^{\mathrm{Id}} - \bar{Y}^{\mathrm{pair}}\right)_{\text{same-ranked}}$$

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- Identification assumption?

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- Identification assumption? Without identity concerns, the difference in take-up between an identity task and its paired control would be the same for every caste
  - "Parallel trends" across castes: anything that makes a task more or less attractive than its pair affects all castes equally, except identity
  - Threat: more willing to do own caste task b/c has more experience with it. (in which case λ^d may be overstated). But experience shouldn't differ between higher and lower tasks, so λ^l is a clean lower bound

Results: identity tasks cut take-up by 23 pp and lower-caste tasks by a further 24 pp

$$Y_{ikt} = \sigma^d D_{ik} + \lambda^d \,\mathrm{Id}_k D_{ik} + \sigma^l L_{ik} + \lambda^l \,\mathrm{Id}_k L_{ik} + \tau \cdot \mathrm{hours}_t + \text{task FE} + \text{worker FE} + \varepsilon_{ikt}$$

	(1)	(2)	(3)
Identity × Different (λ^d)	−0.233	−0.233	−0.244
	(0.043)	(0.044)	(0.047)
Identity × Lower (λ^l)	−0.238	−0.238	−0.205
	(0.031)	(0.032)	(0.037)
Hours on extra task (τ)	−0.055	−0.055	−0.053
	(0.005)	(0.006)	(0.006)
Same-ranked take-up: identity / control	0.722 / 0.717	0.722 / 0.717	0.731 / 0.728
Fixed effects	Task, caste	Task, worker	Task, worker
Task × age, education, wealth			✓
Observations	15,120	15,120	13,224

Is the “identity effect” self image or social image?

- Villages randomized to a public or a private arm (322 public, 308 private workers)
  - Cover story: a village focus group on agricultural practices in the coming days
  - Public: your job choices will be discussed at the focus group, even if you do not attend
  - Private: your choices stay private (except willingness to wash farming tools, a control task)
  - Only who learns the decision changes. The extra task is done in private in both arms

	Take up job offer
Identity × Different (λ^d)	−0.223 (0.059)
Identity × Lower (λ^l)	−0.253 (0.041)
Public × Identity × Different	−0.019 (0.086)
Public × Identity × Lower	0.030 (0.053)

- So is this mostly self image or social image?

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- So is this mostly self image or social image?
  Argues mostly self image (decisions are the same even when private)

Reflections

- People are willing to forego real pay/jobs if they are not identity congruent
  - A bit unclear whether this is all caste identity vs also purity norm about “dirty” tasks?
- Other thoughts?
- Might be different for a permanent job - maybe more willing to do it? Or less willing? Unclear
- Purposefully shut down gender here by design (men only, screen out tasks with gender association), but could be an interesting angle to look at too

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Does financial strain make workers less productive?

- Poor households in India, Bangaldesh, South Africa report constant worry about money (Collins et al. 2009)
- Even in a high-income country like the US, average American reports being distracted by finances 6.4 working hours a week (Sergeyev, Lian, and Gorodnichenko 2023)
- The psychology of scarcity
  - **Mullainathan and Shafir (2013)**: pressing needs capture attention and leave less "bandwidth" for everything else
  - **Haushofer and Fehr (2014)**: poverty causes stress and negative affect, which push toward short-sighted (more present-biased) choices that make it more likely that people who are already poor stay poor
- Does financial stress lower outcomes at work? Could also see argument for increasing effort if money is more needed, so the net effect is unclear

Policy implications of this question...

- **How often to pay workers**: paying more often could raise productivity
  - But with lumpy bills (rent, school fees) and self-control problems, frequent small payments can *raise* strain; what matters is matching pay to expenses
- **Cash transfers**: part of their effect on earnings may run through strain as well as through relaxed credit constraints
  - **Banerjee et al. (2020)**: people given a livestock transfer are more productive in a piece-rate bag-sewing task

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- **Efficiency wages**: firms might pay more to buy focus. Classic reasons are different
  - **Dasgupta and Ray (1986)**: better-paid workers eat enough to work hard
  - **Shapiro and Stiglitz (1984)**: a wage premium makes losing the job costly, so workers don't shirk
  - **Weiss (1980)**: a higher wage attracts better applicants
- **Shocks are amplified**: people are least productive exactly when they most need money

Setting: two-week plate-making jobs in rural Odisha

- Lean season (March–June) of 2017 and 2018: little farm work, and workers find paid work 1.9 days a week

- 408 men aged 18–55 from 47 villages; hired by lottery from those who signed up; 14 rounds of about 30 workers
  - Only men: women could not travel outside the village for work
  - 68% have loans
  - Two-thirds could not easily raise Rs 1,000 (4 days' wages) in an emergency

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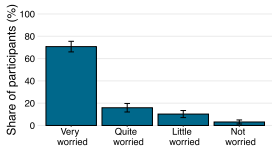
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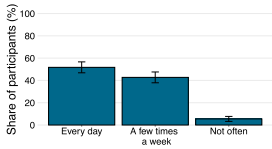
- Contract: 12 days, about 5 hours a day
  - Rs 250 on day 1, then Rs 200/day plus Rs 3 per plate (the piece rate is about 20% of pay)

Workers have high levels of financial worry

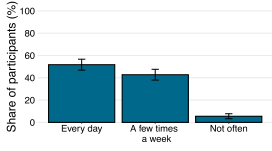
Panel A Intensity of worries



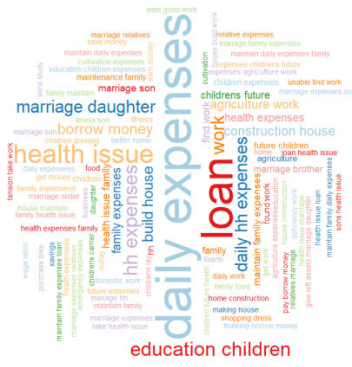
Panel B Frequency of worries



Panel C Duration of worries



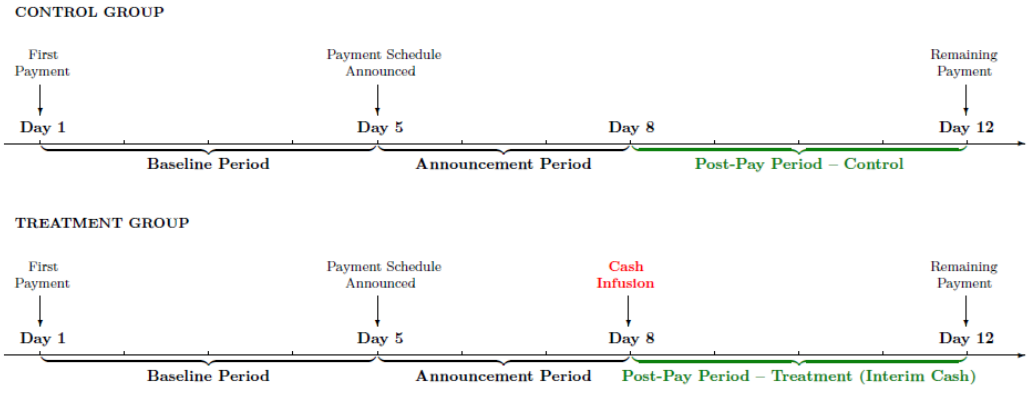
Panel D Triggers of financial worries



Task and outcomes: leaf plates counted every hour

- Stitch sal leaves into disposable plates, sold to local restaurants
  - Each plate needs a plan: fit them into a circle with as few leaves as possible
  - A lapse means unpicking stitches or patching a gap with another leaf, which costs time
- **Output**: accepted plates per hour (over 97% are accepted)
  - Collected every hour to clear the work area and counted in a back room by two staff
- **Attentiveness**: on a subset of days, staff count on every plate
  - Double holes (a stitch removed to undo a mistake), number of leaves, number of stitches
  - Workers don't know these are measured, and they don't affect pay

Design: move the payday while holding pay fixed



Effects of early pay on spending in the interim

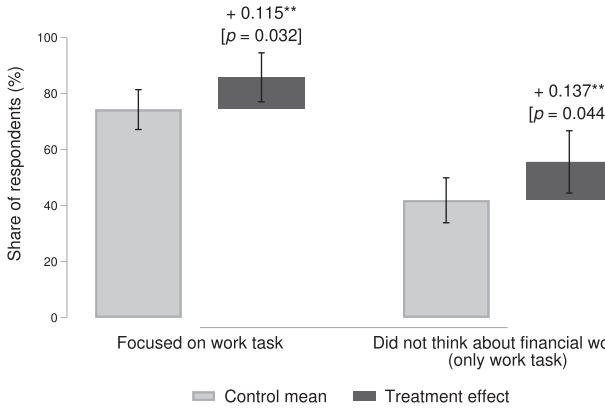
In the 3 days after the interim payment	Effect of cash	Control mean
Loan repayment (Rs)	271*** (54)	94
Repaid any loan	0.40*** (0.04)	0.18
Household spending (Rs)	150*** (39)	372
of which food (Rs)	69*** (24)	270
Tobacco and alcohol (Rs)	−0.28 (4.56)	34
Total spending (Rs)	371*** (68)	568

From a recall survey on the last day

- Most loan repayment happens the same evening the cash arrives
- Spent on loans and household needs (food, clothes, soap, fuel)

Treated workers report thinking less about money while working

- Two days after the interim payment, asked open-ended: “What were you thinking about while you were working today?”
- 60% of controls mention financial or household worries
- Cash group:
  - 11.5 pp more likely to say they were focused on the task
  - 13.7 pp more likely to mention no financial worries
- Ask only once, two days after pay
  - Asking about money every day could itself remind workers of their finances



Estimate the effect of cash on hand on hourly output

$$y_{irdh} = \beta\,\text{Cash}_i\,\text{PostPay}_{ird} + \gamma\,\text{Cash}_i\,\text{Announce}_{ird} + \theta\,\text{PostPay}_{ird} + \mu\,\text{Announce}_{ird} + X_{ir}'\lambda + \delta_r + \varepsilon_{irdh}$$

- y_{irdh}: accepted plates by worker *i* (round-wave *r*) on day *d* in hour *h*, divided by the control group's post-pay SD (1 SD $\approx$ 2.4 plates an hour)
- $\text{Cash}_i = 1$ for the early-pay group.
- $\text{Announce} = 1$ from the day-5 announcement through payday
- $\text{PostPay} = 1$ on days after the interim payment.
- What each control is doing:
  - θ, μ: how everyone's output changes across periods (learning, fatigue)
  - δ_r: round-wave (strata) fixed effects
  - X_{ir}: the worker's baseline output plus baseline controls picked by LASSO
- Coefficients of interest are β and γ. What is β? What is γ?

β and γ are the parameters of interest

$$y_{irdh} = \beta \, \text{Cash}_i \, \text{PostPay}_{ird} + \gamma \, \text{Cash}_i \, \text{Announce}_{ird} + \theta \, \text{PostPay}_{ird} + \mu \, \text{Announce}_{ird} + X_{ir}'\lambda + \delta_r + \varepsilon_{irdh}$$

- β: difference in hourly output between cash and control workers after the cash arrives
- γ: the same difference after the schedule is announced but before any cash arrives
- What do we need for identification?

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- What do we need for identification? Cash & control workers with same baseline output would produce the same in each period if under the same pay timing (should be true b/c randomized)
- What does each story predict for β and γ?
  - Standard model: tiny effects of either
  - Gift exchange: these respond to the *news*, so effects should start at the announcement
  - Financial strain: falls only when cash is in hand, so $\beta > 0$ with $\gamma \approx 0$

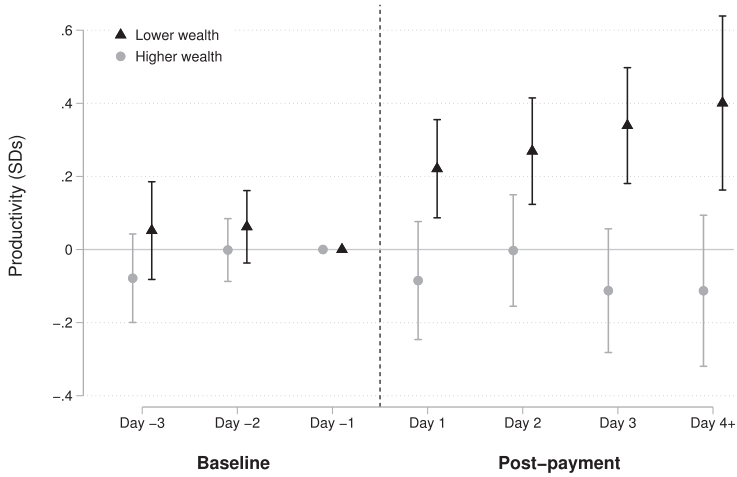
Results: output rises 0.11 SD after the cash arrives and not before

$$y_{irdh} = \beta \, \text{Cash}_i \, \text{PostPay}_{ird} + \gamma \, \text{Cash}_i \, \text{Announce}_{ird} + \ldots$$

	(1) Baseline output	(2) + LASSO controls	(3) + day, hour FE	(4) By wealth
Cash × Post-pay (β)	0.097**	0.109**	0.111**	0.204***
	(0.047)	(0.047)	(0.047)	(0.069)
× Higher wealth				−0.190**
				(0.093)
Cash × Announcement (γ)	−0.002	0.014	0.012	0.039
	(0.035)	(0.035)	(0.035)	(0.061)
p-value: $\beta = \gamma$	.006	.008	.007	.000

0.11 SD is 6.9% more output; 13% for poorer workers, and about zero for richer ones

Poorer workers' output rises the day after pay and stays up



Day 1 is the first day after the interim payment, stacked across the two waves; 90% CIs.

Reflections

- Should employers just pay more often? Not obviously: with lumpy bills and self-control problems, frequent pay could raise strain (the authors raise this)
- Other thoughts?
- All men again

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Do workers care how their pay compares with their co-workers' pay?

- In standard agency models, a worker cares only about her own wage
- Written about by social/behavioral psychologists:
  - **Festinger (1954)** social comparison theory: when there is no objective yardstick, people judge their own abilities and outcomes against similar others
  - **Adams (1963)** equity theory: a worker compares his ratio of outcomes (pay) to inputs (effort, skill) with that of a comparison person

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- Written about by economists:
  - **Akerlof and Yellen (1990)**, the fair wage–effort hypothesis: workers give full effort only if paid at least a "fair" wage that depends on what others earn. Firms then compress pay, and unemployment falls mostly on the low-skilled
  - **Fehr and Schmidt (1999)**, inequity aversion: people lose utility when they earn less than others and, less strongly, when they earn more. Explains fair offers in ultimatum games and punishment in public-goods games
  - **Bewley (1999)**: about 300 interviews, mostly with managers, in the U.S. Northeast during the early-1990s recession. Managers avoid pay cuts and pay gaps among similar workers because they believe these hurt morale

People caring about relative pay might explain several things...

- **Wage compression**: pay varies less than marginal product
  - **Frank (1984)**: inside firms (real estate agents, car salespeople), pay rises much less than one-for-one with productivity. His explanation is *status*: ranking high among co-workers is valuable, so top producers accept less than their marginal product to buy that rank, and bottom producers are paid more to put up with low rank
  - **Drèze and Mukherjee (1989)**: in Indian villages, casual daily wages are strikingly uniform across workers and do not fall in the slack season despite unemployment

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- **Boundaries of the firm**: **Nickerson and Zenger (2008)** argue that envy among employees who compare pay makes it costly to keep very different workers inside one firm, a reason to outsource

Setting: seasonal manufacturing jobs in rural Odisha

- Three workshops near Bhubaneswar set up and run by the researchers
  - 10 products: rope, brooms, incense sticks, candle wicks, leaf plates, mats, paper bags, peanuts
  - Local contractors set quality standards and sell the output in wholesale markets

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  - Local contractors set quality standards and sell the output in wholesale markets
- 378 adult men, 14 one-month rounds, each round hired from a new pair of villages within 15 km
- Flat daily wage, as is the local norm: training wage Rs 250/day plus lunch
  - Fired if more than 3 days absent in a row
  - About 45% have worked under piece rates previously

Each worker's reference group is built into the workshop layout

- On day 1, workers are randomly assigned to production units of 3
  - All 3 make the same product; every unit in a workshop makes a different one
  - Each unit sits in its own area, usually a separate room, and eats lunch there together
  - Production is individual; no joint output

- Creates a natural reference group
  - Your two unit-mates are the only other people making your product: a clear, salient reference group

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- Extra staff collect each worker's output, count it in a back room, and record what is of salable quality
  - Output standardized within task, using the mean and SD of the last 3 training days

Design: co-workers' wages vary while own wage is held fixed

Rank within unit	Pay disparity	Compressed Low	Compressed Medium	Compressed High
Low productivity	w_L	w_L	w_M	w_H
Medium productivity	w_M	w_L	w_M	w_H
High productivity	w_H	w_L	w_M	w_H

- Each unit is assigned one of four pay structures, effective on day 14. Everyone gets a raise over the Rs 250 training wage: $w_L, w_M, w_H \approx$ +4%, +8%, +12% (e.g. Rs 260, 270, 280)

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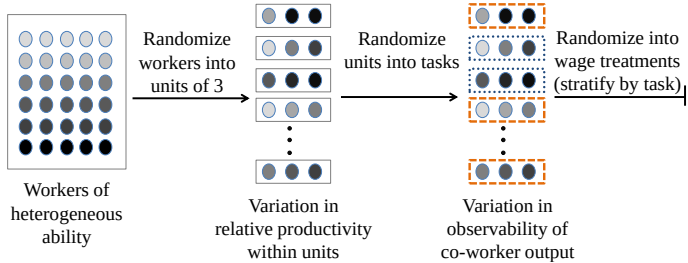
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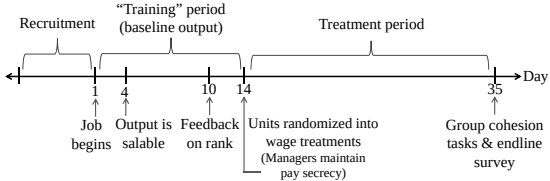
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- Compressed arms against each other: the effect of the *level* of own pay

Three layers of randomization



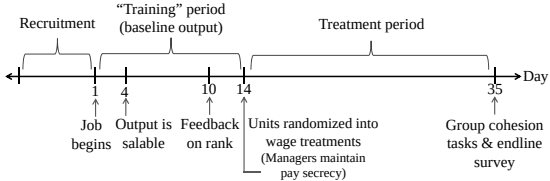
1. Randomize workers into units of 3, creating variation in productivity within group
2. Randomize units:
   - Which product they'll make
   - Whether pay is based on productivity or all set equal (and at what level)
   - Whether their co-workers' output is observable

Timeline: a private wage change after two baseline weeks



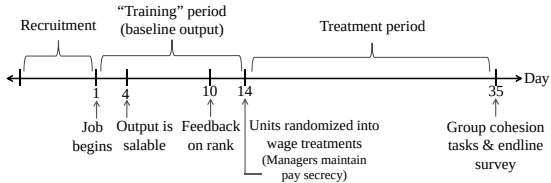
- Day 1: workers into units, units into tasks
- Day 10: private feedback on own rank in the unit.
- Days 11–13: output on these days sets the rank used for pay
- Day 14: each worker told his new wage privately; about 20 more workdays follow
- Day 35: endline survey

Timeline: a private wage change after two baseline weeks



- Day 1: workers into units, units into tasks
- Day 10: private feedback on own rank in the unit.
- Days 11–13: output on these days sets the rank used for pay
- Day 14: each worker told his new wage privately; about 20 more workdays follow
- Day 35: endline survey
- Why tell workers their rank on day 10, *before* the wage change?

Timeline: a private wage change after two baseline weeks



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- Day 35: endline survey
- Why tell workers their rank on day 10, *before* the wage change?
  - So both arms know their rank before pay changes: otherwise only Pay disparity workers would learn it, through the pay gap
  - Day 14 can still update rank (days 11–13 set pay), but ranks are stable, and learning your rank on day 10 had no effect on output

Day 1 script: a one-time training program (again full scripts in appendix)

The goal of this program is to train you on how to make this product. You will be able to participate in this training program for <*insert number*> days. We will end on <*insert day and date*>. After this day, there will be no more work here. No person can work here for more than <*insert number*> days. Our policy is to train each person only once, and only in one product.

To ensure that workers understand, manager should prompt workers with the following questions: So how many days is this work for? What are the different types of products you may be learning to make? How many days will this training program be? When is the last day of work here?

- Delivered by the worksite manager to all new workers; repeated in the daily attendance script
- Comprehension questions are built into the scripts, as in Oh (2023)

Do workers learn their co-workers' pay? (answers from endline)

Share who correctly report …	Compressed	Pay disparity	*p*-value
both unit-mates' wages	0.909	0.742	0.003
at least one unit-mate's wage	0.958	0.871	0.031
all wages in another unit	0.086	0.019	0.000

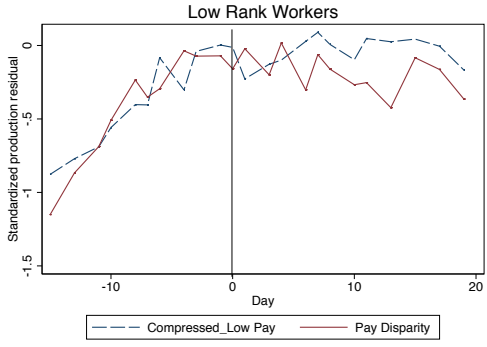
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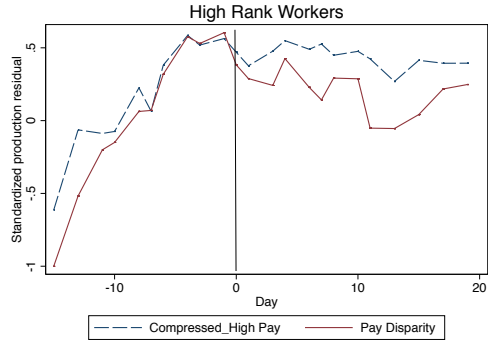
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- Most workers learn their unit-mates' pay; almost no one knows another unit's: the unit is the reference group
- Why might fewer workers know their unit-mates' pay under Pay disparity?
  - The authors' reading: it is awkward to talk about pay when it differs
  - Not just guessing "same as mine": only 4 Pay disparity workers report that both unit-mates earn the same wage

Raw data

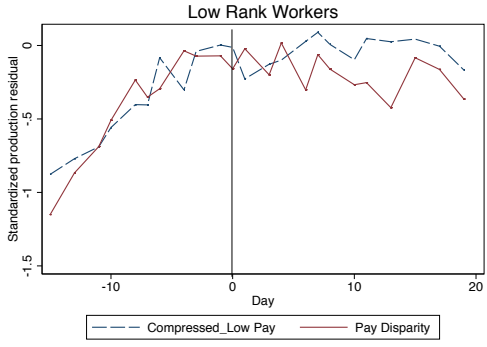
- Left: paid less than both peers. Right: paid more than both peers

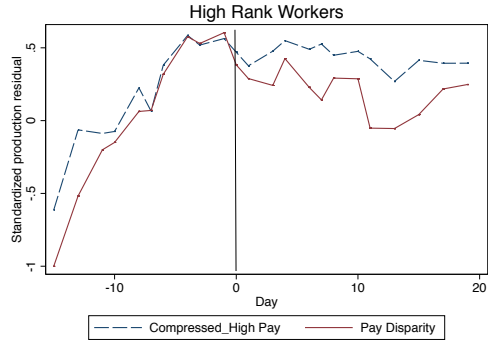




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Output falls in the pay disparity arm once workers learn each other's pay

Estimate the effect of co-workers' pay as a difference in differences

$$y_{it} = \delta_L \, \text{Post}_t \, \text{PayDisp}_i \, \text{Low}_i + \delta_M \, \text{Post}_t \, \text{PayDisp}_i \, \text{Med}_i + \delta_H \, \text{Post}_t \, \text{PayDisp}_i \, \text{High}_i$$
$$+ \textstyle\sum_r \mu_r \, \text{Post}_t \, \text{Rank}_i^r + \lambda_i + \tau_t + \eta_1 x_{kt} + \eta_2 x_{kt}^2 + \varepsilon_{it}$$

- y_{it}: output of worker i on day t (standardized within task, zero if absent), or attendance
- $\text{Post}_t = 1$ after the day-14 wage change
- $\text{PayDisp}_i = 1$ for a pay disparity unit
- $\text{Low}_i, \text{Med}_i, \text{High}_i$: rank in the unit

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- What each control is doing:
  - Worker FE λ_i: baseline productivity, village, anything fixed about the worker or unit
  - Day $\times$ round FE τ_t: shocks common to everyone that day (festivals, rain, paydays)
  - x_{kt}: days of experience, as a quadratic with separate coefficients for each product. Workers are still learning, and broom-making and incense-rolling improve at different rates; day $\times$ round FE can't capture product-specific learning curves
  - μ_r: rank-specific changes after day 14 that happen in both arms. E.g., ranks come from only 3 days of output, so low-ranked workers were partly unlucky on days 11–13 and bounce back (regression to the mean)

The δ's are the parameters of interest

$$y_{it} = \delta_L \, \text{Post}_t \, \text{PayDisp}_i \, \text{Low}_i + \delta_M \, \text{Post}_t \, \text{PayDisp}_i \, \text{Med}_i + \delta_H \, \text{Post}_t \, \text{PayDisp}_i \, \text{High}_i$$
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- δ_L: change in output after day 14 for a low-rank worker in a Pay disparity unit, relative to a low-rank worker in a Compressed Low unit. Both earn w_L, so this is the effect of being paid *less* than both peers
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- δ_M: medium rank (vs. Compressed Medium), paid more than one peer and less than the other

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- Identification assumption?

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- δ_M: medium rank (vs. Compressed Medium), paid more than one peer and less than the other
- Identification assumption? Without the change in pay structure, same-rank workers in Pay disparity and Compressed units would have followed the same path after day 14
  - "Parallel trends" in this case are guaranteed in expectation by randomization
  - So why a DiD at all? For precision... but some (e.g. David McKenzie) say ANCOVA is better

Results: being paid less than both peers lowers output and attendance

$$y_{it} = \delta_L \, \text{Post}_t \, \text{PayDisp}_i \, \text{Low}_i + \delta_M \, \text{Post}_t \, \text{PayDisp}_i \, \text{Med}_i + \delta_H \, \text{Post}_t \, \text{PayDisp}_i \, \text{High}_i + \ldots$$

	Output (SD) (1)	Attendance (2)	Output if present (SD) (3)
Low rank (δ_L): paid less than both peers	-0.332^{**}	-0.120^{**}	-0.204^{*}
	(0.128)	(0.053)	(0.114)
Medium rank (δ_M)	-0.226	-0.129^{**}	-0.061
	(0.187)	(0.060)	(0.138)
High rank (δ_H): paid more than both peers	-0.172	-0.104^{**}	-0.009
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Compressed mean, post period	-0.099	0.939	0.015

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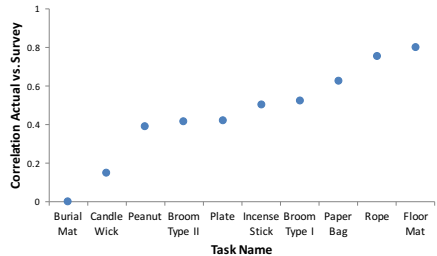
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Negative effects of pay disparity for everyone; no evidence higher-paid worker likes it

How do the results change if co-worker productivity is observable?

- Observability is a property of the *product*, e.g., whether piles of output build up next to each worker
- Measured in a separate pilot before the experiment
  - Different workers on equal pay, never told rank
  - After 3 weeks, each ranks his unit-mates; a product's score is how accurate these guesses are
  - Scores run from 0 to 0.88; "observable" if > 0.5



Pilot workers' accuracy in ranking unit-mates' output, by product

- Observability is **not** randomly assigned
  - Random: which unit makes which product (pay arms stratified by product), so observability is unrelated to the workers or their pay arm
  - Not random: observability itself comes bundled with everything else about the product (difficulty, spread of output, layout of the work)
  - So this is heterogeneity across 10 products, like heterogeneity by region in an RCT

Results: heterogeneity by observability

- Same regression, with Post $\times$ Pay disparity interacted with an observable-product indicator

	Output (SD)	Attendance
Post $\times$ Pay disparity (products that are hard to observe)	-0.384^{***}	-0.153^{***}
	(0.131)	(0.031)
$\times$ Observable product	0.395^{**}	0.100^{**}
	(0.161)	(0.046)

(all ranks pooled)

- What is the effect of Pay disparity on observable products?

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(all ranks pooled)

- What is the effect of Pay disparity on observable products?
  - Output: $-0.384 + 0.395 \approx 0$ SD. Attendance: $-0.153 + 0.100 = -5$ pp
- But this isn't a perfect test.. different task types also bundle other features of the task

Reflections

- Workers care about co-workers' pay, especially when the gap is not visibly earned
- Thoughts?
- Would it be different in a permanent job, where pay gaps also carry dynamic incentives? Larger or smaller? Unclear
- All men again. Would pay gaps by gender or caste look "justified" to workers? (the paper's conclusion raises this)